

AK VisionFlow Owner Android App

Phase 1 — Android Foundation & Secure Read-Only Architecture

1. Phase Objective

Build the foundation of a lightweight Android Owner App that connects securely to the existing AK VisionFlow system. Phase 1 establishes the mobile application architecture, authentication, API connectivity, owner access control, basic navigation, and the read-only security model. No sales, purchase, stock, customer, supplier, invoice, or other operational entry will be possible from the app.

2. Core Product Principle

Web Application = Run the business
Android Owner App = Understand the business

The Android application must remain intentionally lightweight. It will display information supplied by the VisionFlow backend rather than duplicating business logic or operational workflows inside the mobile client.

3. Phase 1 Scope

Module	Phase 1 Deliverable
Android Project	Production-ready Android project structure using Kotlin and Jetpack Compose.
Authentication	Secure owner login, session/token handling, logout, and protected screens.
Owner Access	Dedicated read-only Owner Mobile role/permissions enforced by the backend.
API Layer	Secure connection to VisionFlow backend APIs with standard error handling.
App Navigation	Basic navigation shell for Home, Analytics, Alerts, AI Advisor, and Profile.
Profile	Basic owner profile and account information; logout option.
Read-Only Security	No write permissions for sales, purchases, stock, customers, suppliers, settings, etc.
Performance Base	Lightweight network/data architecture and basic caching strategy.

4. Authentication & Authorization

- Owner logs in using the VisionFlow account credentials.
- Authentication token/session is stored securely on the device.
- Expired/invalid sessions require re-authentication.
- Backend authorization must enforce the Owner Mobile role.
- The mobile app must not rely only on hidden buttons for security; write APIs must reject unauthorized write requests.

5. Mobile Navigation Foundation

Home — executive overview placeholder
Analytics — analytics area placeholder
Alerts — business alerts placeholder
AI Advisor — AI insight area placeholder

Profile — account and app information

Phase 1 creates the navigation structure only. Detailed dashboard statistics, alerts, and AI functionality are implemented in later phases.

6. Backend/API Requirements

- Dedicated mobile authentication endpoint or compatible secure authentication flow.
- Owner profile endpoint.
- Owner permission/role endpoint or token claims.
- Health/status endpoint for API connectivity.
- Standard response/error format for the Android client.
- Versioned API structure so future Android releases can evolve without breaking the web application.

7. Explicitly Out of Scope

No product CRUD, sales entry, purchase entry, stock adjustment, customer/supplier creation, invoice creation, payment entry, expense entry, POS, barcode scanning, printing, branch management, user management, or system settings.

8. Phase 1 Completion Criteria

1. Android app installs and launches successfully.
2. Owner can securely log in and log out.
3. Authenticated screens are protected.
4. Backend correctly identifies the mobile Owner role.
5. Unauthorized write operations are blocked at the API level.
6. Navigation shell works across all five main sections.
7. Basic profile information can be displayed.
8. API/network failures are handled gracefully.
9. Project is structured for Phase 2 dashboard development without major architectural rework.

9. Recommended Technology

Layer	Recommendation
Language	Kotlin
UI	Jetpack Compose
Architecture	Clean/feature-based architecture with clear data, domain, and UI separation
Networking	REST API with secure HTTPS
Authentication	Token-based authentication with secure local storage
Notifications	Firebase Cloud Messaging foundation (detailed in Phase 3)
Backend	Existing AK VisionFlow backend / centralized business APIs

Phase 1 Result: A secure, lightweight Android foundation connected to AK VisionFlow, ready for Phase 2 — Executive Dashboard & Business Statistics.